

An Adaptive and Intelligent Platform for Role-Based Placement Preparation and Readiness Assessment

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***Abstract**—Preparing for job placements on campus is usually done by students using methods that are not connected. These methods include question lists, general practice tests and self-organized interviews. None of these methods are tailored to a job role or to the students own skills and areas that need improvement. Traditional systems usually give one test and do not consider how the student did before. This leads to the questions being seen again and again. It also gives information about which subjects need more work. This paper introduces a web-based platform that helps students prepare for job roles. The system lets a student choose a target job practice for that job and take a test with types of questions. The system chooses questions randomly. Does not repeat ones the student has already seen. After the test the system looks at the students performance in topics and finds the weak areas. If the student does not meet a set score the system offers a chance to try again focusing on the weak areas. When the student passes the platform gives access to an AI-based interview module. This module takes text or voice answers changes voice to text. Gives feedback on things like how relevant the answer is, how correct it is, how clear it is, how complete it is and how professional it sounds. There is also a way to track progress and a measure of how ready the student's for placements. The paper focuses on the process of preparation rather, than just a single machine-learning tool. It also clearly separates the AI parts from the engineering parts.*

***Index Terms**—Adaptive Learning, Placement Preparation, Role-Based Assessment, Personalized Assessment, Artificial Intelligence, Large Language Models, HR Interview, Speech Recognition, Text-to-Speech, Student Performance Analysis, Weak Topic Detection, Placement Readiness*

I. INTRODUCTION

Campus placement preparation has become a task for undergraduate students. The reason is that recruiters are now expecting technical skills, especially in software development, data analytics and machine learning. Students usually depend on resources to get ready—like question banks, video tutorials coding websites and mock interviews arranged by peers. These tools can help on their own. They don't work well together. They also rarely match the job role a student wants.

One big problem with preparation materials is that they are not role-specific. Whether a student is going for a Python developer job, a data analyst role or a machine learning engineer position they often face the general questions.. In real hiring each role has different expectations. The topics emphasized vary. So do the interview styles. A Python developer might be tested heavily on frameworks and syntax while a data analyst may need stats and visualization knowledge. Yet current resources treat all candidates the same.

Another issue is that question banks are static. Once a set of questions is made students who take the test again usually see the same questions or ones very similar to them. This makes repeated practice less useful. Of helping identify what a student still struggles with it just lets them memorize answers. That's not how learning works.

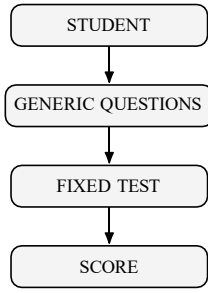


Fig. 1. Traditional, role-agnostic placement preparation workflow.

Assessment performance is used to diagnose areas that need focused assessment. The diagnosis is then translated into an assessment that is more targeted.

Beyond preparation interview readiness represents another gap. Human resources and behavioral interview preparation are often treated as an activity. This activity is disconnected from a students demonstrated performance. Moreover it is rarely subjected to any form of repeatable feedback. Recent progress in natural language processing, automatic speech recognition and text-to-speech synthesis has made it feasible to simulate an interview experience. Structured feedback also allows the generation of feedback, on a candidates response. Such feedback carries the limitations associated with automated language understanding.

Figure 1 illustrates the workflow. In this workflow a student is presented with questions and a fixed test. The result is a score and no further diagnostic feedback. By contrast the workflow proposed in this paper is summarized in Figure 2. It extends preparation into an iterative role-specific cycle. This cycle connects assessment performance to analysis, adaptive reattempt, qualification-gated interview practice and progress tracking.

This paper introduces a placement preparation platform. The central purpose of this platform is to convert assessment results into a adaptive preparation process. The platform is not a terminal score. The remainder of this paper is organized as follows. Section II reviews related work. Section III identifies the research gap and problem statement. Section IV summarizes contributions. Section V describes the system modules. Section VI presents the architecture. Section VII details the methodology. Section VIII describes the workflow. Section IX discusses database design. Section X outlines methodology.

II. RELATED WORK

A. Traditional Placement Preparation

Conventional placement preparation is mostly built around question banks, printed reference material and periodic mock tests that are given without adaptation to each student's performance [6]. Conventional placement preparation offers topic coverage yet it does not personalize question selection to a student's demonstrated ability. Conventional placement preparation also separates assessment, from interview preparation treating technical assessment and interview preparation as unrelated activities.

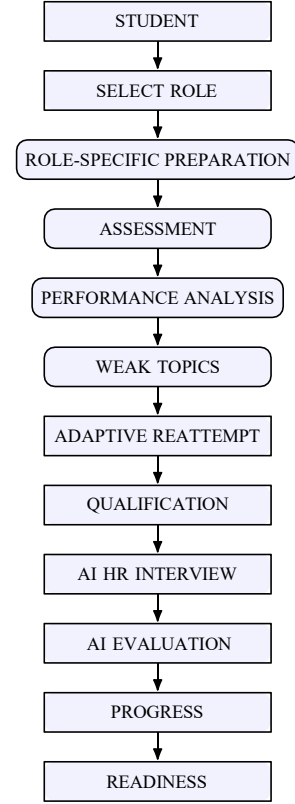


Fig. 2. Proposed adaptive, role-specific placement preparation workflow.

B. Adaptive Learning and Knowledge Tracing

Adaptive learning systems try to understand how a learners knowledge changes over time and then change what comes next. Early research in knowledge tracing looked at the chance that a student has learned a skill by looking at the history of their practice attempts [1]. This idea was later improved by using deep learning methods called neural networks in something called Deep Knowledge Tracing [2]. Brusilovsky created an important way to make hypermedia systems adapt to each user by using a model of the person [3]. Studies, on technology that helps make learning personal show that most systems change either the order of content or how hard it is.. Very few of them use personal assessments and then have a structured interview after that [12].

C. Computer-Based and Automated Assessment

I have read that educational data mining research looks at how digital assessment logs can be used to model student performance and student behavior [5]. In automated and computer-based assessment systems the system uses randomized item selection from a question pool to make the test less predictable. However randomization alone does not ensure that a student who tries again will get a different set of items unless the system removes items that the student has already answered.

D. AI in Education

AI techniques have grown a lot in education after large language models improved. AI techniques are now a part of teaching.

I find that the Transformer architecture [9] forms the base of large language models such as BERT [7] and GPT-3 [8] and these models let us create natural-language feedback that's more flexible than the older rule-based tutoring systems. Intelligent tutoring systems, reviewed by VanLehn [4]. Described in depth by Woolf [10] demonstrated that structured feedback-driven tutoring can approach the effectiveness of tutoring in constrained domains. Recent surveys of AI-enabled learning systems report substantial interest in personalized content delivery but note that end-to-end integration, with placement-oriented outcomes remains limited [11].

E. AI-Based Interview and Speech Systems

Automated interview analysis has been explored using both modal and language-based methods. Naim et al. Developed a system that automatically analyzes and predicts job interview performance by looking at nonverbal signals [13]. Hemamou et al. Created an attention model to evaluate asynchronous video job interviews [14]. Rasipu-ram and Jayagopi looked into assessing communication skills during interview-based interactions [15]. On the speech processing side deep learning was used for scale automatic speech recognition by Hannun et al. [16]. By Graves et al. Using recurrent networks [17]. Modern neural text-to-speech synthesis was shown by Shen et al. In Tacotron 2 [18]. These studies together show that voice-based interview simulation is technically possible.. They also point out that the results can be sensitive to audio quality, accent and differences, in the topic or domain being discussed.

F. Student Performance Prediction

Learning analytics research has looked into predicting student performance and finding students who're, at risk using data from their behavior and test results [5] [6]. This kind of work supports the idea of analyzing how well students do on topics to help plan better interventions. In this paper that idea is adjusted for use in the context of placement preparation.

I have seen that existing studies mainly look at one part of learning, assessment or automated interviewing by itself. A workflow that ties together role-specific assessment, adaptive reattempt and AI-based interview evaluation all, in one placement-preparation platform is not as well studied in the literature.

I. RESEARCH GAP AND PROBLEM STATEMENT

A. Research Gap

Conventional preparation systems follow the workflow shown in Fig. 3. Conventional preparation systems rely on a fixed question bank that produces a fixed assessment and a terminal score.

Adaptive-learning systems usually take this idea further into a performance-driven personalization loop as shown in Figure 4. However Adaptive-learning systems usually stop at content personalization. Do not extend toward interview readiness.

The research gap this paper focuses on is how to bring role-specific preparation, adaptive assessment that reduces

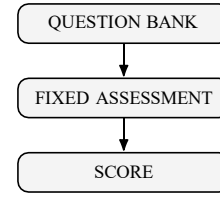


Fig. 3. Conventional fixed-assessment workflow.

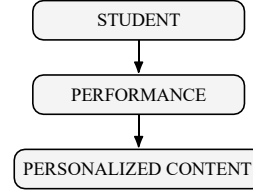


Fig. 4. Generic adaptive-learning personalization loop.

repeated questions, analysis of weak topics AI-based interview simulation and progress tracking all into one clear and unified placement-preparation workflow.

B. Problem Statement

(i) The system must offer practice material tailored to the role; (ii) The system must build a test for the role but must not include questions the student has already answered; (iii) The system must check the student's replies and give a score; (iv) The system must highlight topics where the student answered wrong; (v) The system must provide a new test that adapts to the student if the score is below the required level; (vi) The system must decide if the student meets the role's requirement; (vii) The system must open an AI-based HR interview when the student meets the requirement; (viii) The system must judge the interview answers; (ix) The system must keep a record of the student's progress each time; (x) The system must estimate how ready the student is, for placement.

The main assessment relationship can be written like this:

$$f(R, Q, A) \rightarrow S \quad (1)$$

Here R stands for the chosen role. Q refers to the set of questions given to the student. A means the set of answers, from the student. S represents the assessment score that comes out of the process.

II. RESEARCH CONTRIBUTIONS

I think this paper makes a difference, between technical/AI-oriented contributions and system/application-level contributions because not every feature of the platform is a new algorithmic contribution.

1) Technical / AI-Oriented Contributions:

- A way to choose questions that changes based on the role a person has focuses more, on topics that're not strong and leaves out questions that have already been tried.
- A weak-topic analysis mechanism that measures topic-level performance from responses.
- A clear and detailed AI-based evaluation methodology examines HR interview responses, for relevance, correctness, clarity, completeness and professionalism.

TABLE I
PROPOSED SYSTEM COMPONENTS

Component	Function	Input	Output
Authentication	Identity and session	Credentials	Session, profile
Role selection	Maps role to skills	Role choice	Topic set
Practice	Optional preparation	Topic/format	Feedback
Assessment	Role-specific eval.	Qs, answers	Score
Adaptive engine	Weak-topic reattempt	Score, topics	New Qs
AI interview	HR simulation	Voice/text	Evaluation
Speech proc.	STT/TTS conversion	Audio/text	Text/audio
Progress	Historical tracking	Attempt history	Progress view
Readiness	Internal indicator	Aggregated metrics	Score

- Putting speech-to-text and text-, to-speech together in a system that simulates interviews and knows the role it is playing.

2) System / Application-Level Contributions:

- A system that uses roles to help prepare people for jobs and supports eight different areas of technical work.
- A mixed-format assessment framework that supports choice questions multiple select questions, output prediction tasks, pseudocode exercises and coding questions; the assessment framework is designed for flexibility.
- A way to choose questions that never repeats for each student.
- A qualification mechanism that ties technical assessment performance to interview access.
- I find the progress tracking module because it mixes technical assessment, with interview performance.

This paper does not say that each feature listed is a machine-learning algorithm. Some parts, like authentication, role selection and the database schema are engineering work that helps the adaptive workflow overall.

III. PROPOSED SYSTEM

The platform is made up of connected parts: authentication and profile management choosing a role optional practice, assessment for each role adaptive reattempt, qualification logic, AI-driven HR interview tracking progress and being ready for placement. Table I shows these parts along, with their inputs and outputs.

A. Authentication and Role Selection

A student signs up. Logs in before they can go to the dashboard. The job role that is chosen then decides which skills are needed which topics are covered, which practice questions are available which assessment questions are used, which areas of weakness are checked and what the interview situation is like. Looking into the role, on its own does not start an assessment or give a score; if no test has been done for the role the status is clearly shown as "Not Attempted". There is no number score.

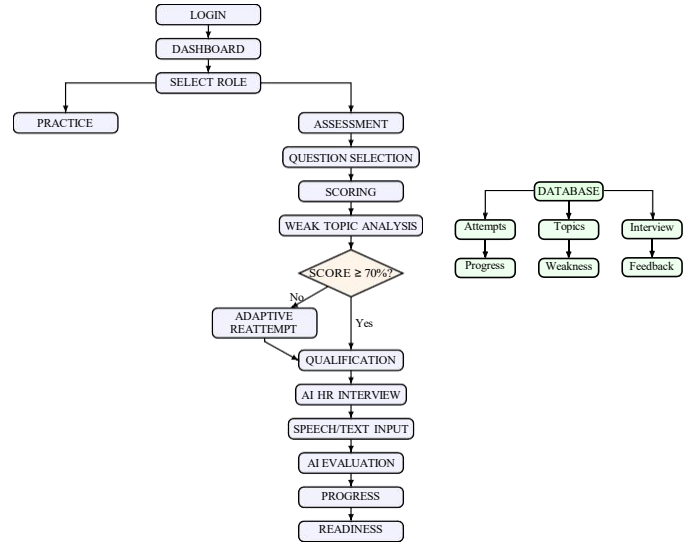


Fig. 5. Overall system architecture of the proposed platform.

B. Practice

Practice is optional. Can include multiple choice questions, multiple select questions, output prediction, pseudocode and coding questions. Practice attempts are kept separate from assessment attempts. Do not affect qualification. This ensures a difference, between casual preparation and the formal assessment that matters for qualification.

C. Role-Specific Assessment

Each supported role is linked to a topic focus. For instance a Python Developer assessment might focus on Python syntax, functions, object-oriented programming, exception handling, data structures and algorithmic problem solving. A Data Analyst assessment could emphasize SQL, Python, statistics and data interpretation. Meanwhile a Machine Learning Engineer assessment may highlight Python machine-learning concepts, model evaluation and data preprocessing.

IV. SYSTEM ARCHITECTURE

Figure 5 shows the system design. It illustrates the process starting with authentication. Then it moves to role selection. Next comes practice and evaluation. After that is reattempt. Then qualification takes place. Following that is the AI interview. Finally there is progress and readiness calculation. All of this is supported by a database.

V. METHODOLOGY

A. User Authentication and Profile Management

The system has sign up log in log out and manages sessions with areas. A profile keeps the students name, email, user type which's either student or an authorized administrator and an authentication code. Protection, at the route level makes sure that the assessment, interview and progress pages can only be seen by people who are logged in.

B. Role Selection

Role selection establishes a mapping

$$\text{Role} \rightarrow \text{Skills} \rightarrow \text{Topics} \rightarrow \text{Questions} \quad (2)$$

This set of rules decides what will be covered in the practice, assessment and interview parts. The eight roles that are covered are Python Developer, Java Developer, Data Analyst, Data Scientist, Machine Learning Engineer, Web Developer, Full Stack Developer and Software Developer.

C. Assessment Construction

An assessment is built by filtering the question bank based on role and topic. Then a suitable mix of question types—, like choice questions (MCQ) multiple select questions (MSQ) output prediction, pseudocode and coding—is chosen. Randomization is. No-repeat rules are enforced, as explained in Sections VII-D and VII-E.

D. Question Randomization

For a role-specific question pool Q , an assessment set Q' of size N is generated as follows:

$$Q' = \text{RandomSample}(Q, N). \quad (3)$$

E. No-Repeat Question Selection

For a given student Q is the set of all questions for the selected role and A is the set of questions that the same student has already tried. The pool of questions, for a new attempt is

$$Q_{\text{available}} = Q - A, \quad (4)$$

and a new assessment is generated as

$$Q_{\text{new}} = \text{Sample}(Q_{\text{available}}, N). \quad (5)$$

F. Assessment Scoring

For an assessment, with N questions each question has a weight w_i and a correctness value c_i that can be 0 or 1 the score is calculated as

$$S = \frac{\sum_{i=1}^N w_i c_i}{\sum_{i=1}^N w_i} \times 100. \quad (6)$$

The qualification threshold is configured as

$$S \geq 70, \quad (7)$$

In this setup 70 serves as the default value and the system can change 70.

G. Weak Topic Identification

After a submission the system checks the answers for each topic. For a topic t with C_t answers, out of N_t attempted questions the performance of the topic is

$$P_t = \frac{C_t}{N_t} \times 100. \quad (8)$$

Topics that have P_t values are flagged as weak areas. The low P_t values, in these topics always signal an area Adaptive

Reattempt

When the score S is than 70 the student is marked as not qualified. The system then supports a reattempt. This process does two things. First it analyzes the answers given by the student. Second it identifies the topics using method (8). It also excludes any questions that the student has already attempted following rule (4). Then the new assessment includes a representation of questions, from the weak topics. Finally the system generates a score based on this updated test..

H. AI-Based HR Interview

Upon qualifying the student can take an AI-based HR interview. The questions are chosen from a set of behavioral questions. Figure 6 shows the voice-interview process.

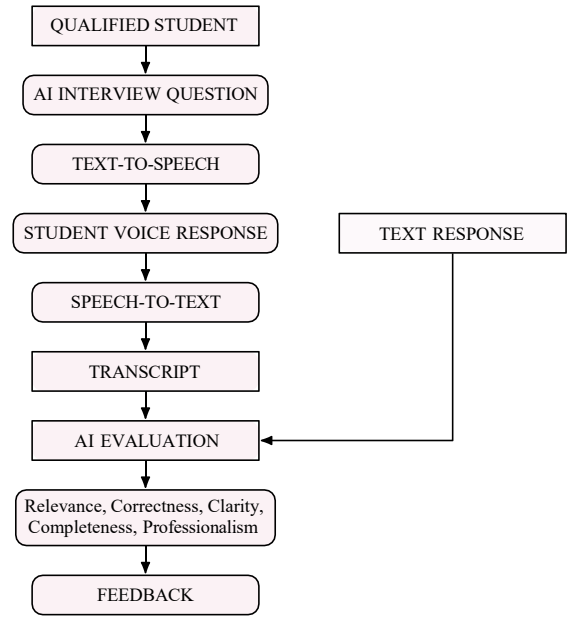


Fig. 6. AI HR interview workflow with voice and text fallback.

I. AI Interview Evaluation

I evaluate interview responses by looking at five aspects: relevance, correctness, clarity, completeness and professionalism. I label each as R , C , L , K and P . Then combine these five aspects into a single score.

$$I = \alpha R + \beta C + \gamma L + \delta K + \epsilon P, \quad (9)$$

where α , β , γ , δ , ϵ are configurable weights.

J. Progress Tracking

The progress module keeps track of how times someone has taken an assessment the scores they got the topics they need to work on the practice activities they did how they did in interviews and what they did every day. This is done for each role. The improvement, between two tries is shown as

$$\Delta S = S_{\text{new}} - S_{\text{old}}. \quad (10)$$

K. Placement Readiness

An internal readiness indicator is proposed as

$$R_d = \lambda_1 T + \lambda_2 P + \lambda_3 I + \lambda_4 C + \lambda_5 A, \quad (11)$$

Algorithm 1 Adaptive Placement Assessment and AI Interview Workflow

```

1: Authenticate student and load profile
2: Student selects placement role  $R$ 
3: Load role-specific topic set and question pool  $Q$ 
4: if student requests practice then
5:   Provide optional, non-qualifying practice content
6: end if
7: Retrieve previously attempted questions  $A$  for student,  $R$ 
8:  $Q_{\text{available}} \leftarrow Q - A$ 
9:  $Q' \leftarrow \text{RandomSample}(Q_{\text{available}}, N)$ 
10: Present assessment  $Q'$ ; collect student responses
11: Compute score  $S$  using Eq. (6)
12: Compute topic performance  $P_t$  using Eq. (8)
13: Identify weak topics  $W \leftarrow \{t : P_t < \tau\}$ 
14: if  $S < 70$  then
15:   Mark student as Not Qualified
16:   Generate adaptive question set emphasizing  $W$ , excluding  $A$ 
17:   goto step 9 (adaptive reattempt)
18: else
19:   Mark student as Qualified for role  $R$ 
20:   Unlock AI HR interview for role  $R$ 
21: end if
22: for each interview question  $q_j, j = 1, \dots, 5$  do
23:   Convert  $q_j$  to speech (text-to-speech)
24:   Capture student response (voice or text)
25:   if response is voice then
26:     Convert speech to text
27:   end if
28:   Evaluate using Eq. (9); store feedback
29: end for
30: Update progress and readiness indicator  $R_d$ 

```

where T refers to assessment performance, P refers to practice performance, I refers to interview performance, C refers to consistency or progress A refers to activity and $\lambda_1 \dots \lambda_5$ refer to configurable weights.

VI. ALGORITHM AND SYSTEM WORKFLOW

Algorithm 1 shows the end-, to-end assessment and AI interview workflow.

Fig. 7 Shows the adaptive-assessment feedback loop by itself. Fig. 8 Shows the student-facing journey starting from login and going through readiness.

VII. DATABASE DESIGN

The relational schema contains the entities profiles, roles, skills, topics, questions, question_options, practice_attempts, assessment_attempts, assessment_answers, coding_submissions, interview_attempts, interview_answers, interview_feedback and activity_sessions. Each profile is linked to a selected role to zero or more practice_attempts and assessment_attempts and to interview_attempts; each assessment_attempt is linked to assessment_answers to a score and, to derived weak_topics..

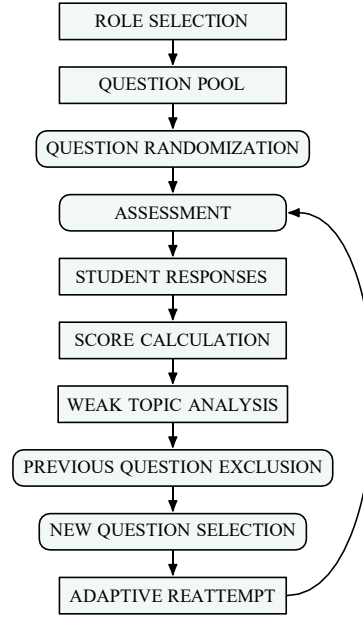


Fig. 7. Adaptive assessment workflow with feedback loop.

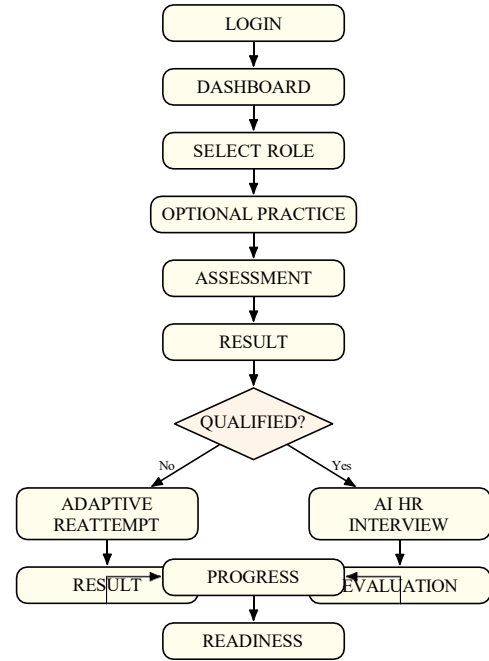


Fig. 8. Complete student user journey from login to placement readiness.

An assessment attempt record is conceptually represented as

$$A_i = (ID_i, U_i, R_i, Q_i, S_i, T_i), \quad (12)$$

and a question record is represented as

$$Q_i = (ID_i, R_i, T_i, D_i, F_i). \quad (13)$$

A. Security and Data Isolation

Frontend-level checks alone are not enough, for access control. The system depends on server-side authentication, authorization and row-level security to make sure a student

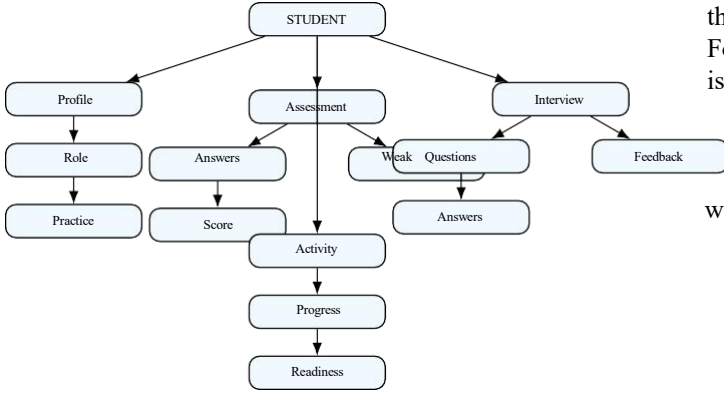


Fig. 9. Database entity relationships and progress/readiness flow.

cannot see another student's assessment answers, scores, interview responses, progress or profile. The service-role database credentials are never shared with the client.

VIII. EXPERIMENTAL METHODOLOGY

A. Implementation Stack

A sample implementation stack includes Next.js and React for the web application TypeScript for type- development Tailwind CSS for styling Supabase Auth for authentication, PostgreSQL as the relational database, an LLM-based service for interview question generation and evaluation speech-to-text and text-to-speech services for the voice interview pipeline, the Monaco Editor for the coding-assessment interface and Git/GitHub, for version control.

B. Dataset and Question Repository

The new platform does not use a public benchmark dataset; instead it needs a organized question database that is specific to each role. Each question in this database has metadata. The metadata includes question identifier, role, skill, topic, difficulty, question type, correct answer, explanation and weight. Table II shows this database as an example. The exact number of questions, for each role depends on how it's set up and is not made up here.

TABLE II
ASSESSMENT REPOSITORY TEMPLATE

Role	Count	Topics	Status
Python Developer	TBD	Python, OOP, DSA	Template
Java Developer	TBD	Java, OOP, DSA	Template
Data Analyst	TBD	SQL, Statistics, Py	Template
Data Scientist	TBD	Statistics, Python, ML	Template
ML Engineer	TBD	ML, Python, Alg.	Template
Web Developer	TBD	HTML, CSS, JavaScript	Template
Full Stack Dev.	TBD	Frontend, Backend, DB	Template
Software Dev.	TBD	Programming, DSA, OOP	Template

C. Evaluation Metrics

Three categories of evaluation are proposed. Assessment-level evaluation includes score-correctness verification, question-selection correctness, adaptive-selection correctness,

the no-repeat rate, and weak-topic identification consistency. For adaptive-assessment evaluation, the question-repetition rate is defined as

$$\text{RepeatRate} = \frac{N_{\text{repeated}}}{N_{\text{total}}} \times 100, \quad (14)$$

weak-topic coverage in a reattempt we define it as:

$$W_c = \frac{N_{\text{weak}}}{N_{\text{adaptive}}} \times 100. \quad (15)$$

IX. RESULTS AND DISCUSSION

A. Functional Validation

Functional validation is planned for areas. Functional validation covers authentication, role selection, practice, assessment submission and scoring weak-topic identification, adaptive reattempt, qualification-gated interview unlocking and progress/readiness updates. Table III shows the metrics that will be filled with data, from an experimental deployment.

TABLE III
EVALUATION METRICS TEMPLATE

Metric	Value
Assessment Score Accuracy	TBD
Question Repetition Rate	TBD
Weak Topic Coverage	TBD
Adaptive Improvement	TBD
Interview Evaluation Agreement	TBD
Assessment Response Time	TBD
Interview Processing Time	TBD

B. Adaptive Assessment Evaluation

Adaptive-assessment behavior is expected to be evaluated using Eq. (14). Eq. (15). These equations are computed over logged attempt sequences. The evaluation compares the no-repeat selection method with a weak-topic-weighted selection approach. It also compares these methods against a random baseline selection.

TABLE IV
COMPARISON OF EXISTING APPROACHES

Approach	Role Based	Adpt. Asmt.	No- Rpt.	Weak Topic	AI Intv.	Prog.	Read.
Fixed bank [6]	No	No	No	No	No	No	No
Know. tracing [1], [2]	No	Yes	Prt.	Yes	No	Prt.	No
Adpt. hyper. [3]	Prt.	Yes	No	Prt.	No	No	No
Intel. tutor. [4], [10]	Prt.	Yes	No	Yes	No	Prt.	No
Auto. intv. [13], [14]	No	No	No	No	Yes	No	No
Proposed	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table IV is constructed from the general capabilities reported in the cited literature. It illustrates that the reviewed individual systems typically address a subset of the listed capabilities, whereas the proposed workflow is designed to address all of them jointly.

C. System-Level Evaluation

System-level evaluation aims to check how fast things respond how long it takes to get data from a database. How long it takes to process interviews when people are using the system. Table V shows a sample situation that helps explain the idea.

TABLE V
ILLUSTRATIVE CONCEPTUAL WORKFLOW SCENARIO

Field	Illustrative Value
Selected Role	Python Developer
Assessment Result	3% (illustrative)
Status	Not Qualified
Weak Topics	OOP, Exception Handling, Data Structures
Adaptive Reattempt	Different questions emphasizing weak topics
Later Result	Use actual value if available
Qualification	Based on configured threshold
AI Interview	Unlocked only after qualification

D. State Management Discussion

Correct behavior of the platform depends on the ability to tell apart application states such as `role_selected`, `assessment_not_started`, `assessment_submitted`, `result_available`, `not_qualified`, `qualified`, `interview_unlocked` and `interview_completed`. When the platform moves incorrectly between these states, it can give wrong scores, progress records, wrong readiness values, or let users enter interviews too early. That is why it is essential to manage these states; it is a core requirement and not a cosmetic detail.

X. ADVANTAGES AND LIMITATIONS

A. Advantages

The proposed workflow provides role- preparation and personalized assessments in mixed formats. It generates assessments randomly which helps reduce the chance of seeing the same questions again. It identifies topics at a topic level and uses that information to create an adaptive reattempt system that focuses more on difficult areas. The system includes role- qualification checks to ensure users meet requirements. It supports AI-based HR interviews that work with both voice and text giving clear feedback. Progress in both skills and interview performance is tracked together. There is also a customizable placement-readiness indicator. All of these features are built into one workflow avoiding the need, for separate tools.

B. Limitations

The platform works well only if the question repository is good and wide. The platform needs a good question repository for results. Wrong topic tagging hurts topic tagging, weak-topic analysis and adaptive selection. The platform cannot do much when the question repository is small because randomization and no-repeat exclusion lose power. AI-based interview evaluation is limited by the problems as any AI-based natural language understanding. Automatic speech recognition is sensitive, to background noise, accent and pronunciation. The platform cannot promise that the internal readiness indicator will lead to job placement.

XI. FUTURE SCOPE

Future work could expand the question repository to a larger set, validated by experts with difficulty levels assigned through peer review. It could also include knowledge-tracing models to improve how questions are selected adaptively. We can model how skills develop over time across attempts and different learning sessions. The coding evaluation system could be made stronger, by using execution with more comprehensive test cases. We could build datasets of interview responses that are annotated by humans to help test and validate the feedback generated by AI. Speech recognition could be made reliable and accurate. Finally the readiness indicators could be fine-tuned by calibrating their weights based on real-world results over time..

XII. CONCLUSION

This paper introduced a platform that helps people get ready for placement by adapting to their needs. The platform links tests that are specific to each role with analysis of weak areas the chance to try again practice for interviews using AI and support that shows how much someone is improving. Of saying the platform is a new machine-learning discovery the paper clearly separated the parts that are about AI technology like choosing the right questions finding weak areas and evaluating interviews with AI from the parts about how the system works and how it is used. The main idea is that the platform changes the way people are tested for a job. It makes the testing process adjust to each person and keep connecting their performance with ways to get better and be ready for interviews. It also makes sure not to promise that the platform can predict results or guarantee a job. More tests are needed to prove that this new way of testing works better, than tests or random ones using the methods described in this paper.

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